

3 minor loop compensation

Given $\epsilon = 10^{-5}$ and plant $P(s) = \frac{1}{s+1}$

a) $H(s) = Ks$, this represents feedback, which increase damping without increasing the system type or e_{ss}

b) for $\omega_c = 10^4$ and $\phi_m = \pi/5$, $G_{inner} = \frac{10^4}{5 \times 10^4 Ks}$

solving for K to satisfy the crossover at $\omega_c = 10^4$

c)

$$E_R = \frac{1}{1 + G(j\omega) H(j\omega)}$$

$$d) \left| \frac{E}{R} \right| = \frac{1}{1 + L(j\omega)} \quad \epsilon \times 10^{-5} \rightarrow (L(j10^4)) \gg \omega \gg 0$$

$$\omega_c = 10^4$$

$$\longrightarrow (10^{-5}s + 1)^2 \approx 1 \longrightarrow \text{low frequencies mean}$$

very little $H(s)$

$$\sim \frac{10^5}{1 + (H(j10^4))} \gg \omega \gg 0$$

option 1: ✓

$$H \approx \frac{10^{-9} (10^9)}{1} = -1 \approx -1 \rightarrow \text{large } k, \text{ and}$$

very little e_{ss}

option 2: X large e_{ss}

option 3: X $|Z| = \frac{10^8}{2r} \approx 1000$, $1000 \neq 10000$

// 4: X like option 2

// 5: X $|Z| = \frac{10^8}{111} \approx 909$, $909 \neq 10000$

// 6: ~~2~~ X like option 2

e)

$$H(s) = \frac{10^{-9} r}{10^{-9} s + 1}$$

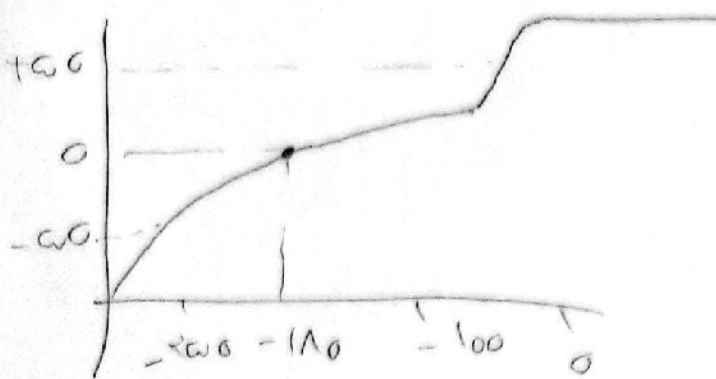
$$r = 10^{-9}, \quad w_c = 10^9$$

$$\Rightarrow L(s) = \frac{(10^9)}{H(s) + (10^{-9} s + 1) r}$$

$$\longrightarrow H(j10^9) = \frac{1}{1 + 0.1j}$$

L) TCP network link

a)



$$M(s) = \frac{\omega \lambda \tau_0}{s + \omega \lambda \tau_0}$$

$$L=1 \rightarrow G(s) = \frac{V \tau / V_0 e^{-\frac{1}{T_0} s}}{(s + \tau_0)(s + \tau)(s + \omega \lambda \tau_0)} = P(s) M(s)$$

ω_{pc} :

$$\phi_m(\omega) = -\arctan\left(\frac{\omega}{\omega \lambda \tau_0}\right) = -(\omega / \omega) \lambda \tau_0 \frac{1}{\tau_0}$$

$$\rightarrow \angle G(j\omega) = -\left[\tan^{-1} \frac{\omega}{\omega \lambda \tau_0} + \tan^{-1} \frac{\omega}{\tau_0} + \tan^{-1} \frac{\omega}{\tau} \right] - \phi_m(\omega)$$

$$|G(j\omega)| = \frac{V \tau / V_0}{\sqrt{\omega^2 + \tau_0^2} \sqrt{\omega^2 + \tau^2} \sqrt{\omega^2 + (\omega \lambda \tau_0)^2}}$$

$$\Rightarrow \phi(V/\tau) = -\left[\tau_0 + \omega \tau_0 \tau + \tau + V \tau_0 \right] = -\tau \tau_0$$

$$PM = \angle G(j\omega) + 180^\circ = -\omega / \tau_0$$

negative phase margin means unstable system ✓